

Enterprise Cloud Storage Sustainability & Cost Audit Report

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Target Storage Scope:	Multi-Cloud AWS S3, Google Drive & Local Filesystem
Compliance Standard:	ISO 14001 / Green IT Cloud Standard

Executive Summary: This audit report provides a comprehensive analysis of unstructured and dark data residing within the organization's cloud storage footprint. Leveraging machine learning classification models, this system identifies redundant, obsolete, and trivial (ROT) file structures to minimize cloud operational expenditure and drastically curtail unnecessary server carbon footprints.

1. Key Performance Indicators (KPI Summary)

TOTAL FILES AUDITED	DARK DATA RATIO	CURRENT CARBON	ESTIMATED MONTHLY ROI
1,000	41.2%	24342.5 g	\$0.64 USD

2. Environmental Impact & Sustainability Breakdown

Continuous hosting of unaccessed dark data accounts for **24342.50 g CO₂** in active emissions. By executing automated migration to cold storage and purging unneeded files, the organization can reduce greenhouse gas emissions by **11475.90 g CO₂** annually.

Metric Description	Value	Environmental & Financial Impact
Active (Useful) Files Count	588	Optimal utilization of storage resources
Dark Data (Unused) Files	412	Wasting cloud power & recurring costs
Prevented CO ₂ Emissions	11475.90 g CO ₂	Equivalent to 14 tree seedlings absorbing CO ₂
Potential Cost Reduction	\$0.64 / mo	Direct AWS / Cloud infrastructure cost savings

3. Strategic Action Plan & Recommendations

- 1. Execute Immediate Cold Storage Migration:** Transition unaccessed files older than 180 days to AWS Glacier Deep Archive to trim active storage costs by up to 75%.
- 2. Enforce Automated Cleanup Policies:** Enable 90-day retention policies on temporary and log files flagged with over 90% AI confidence.
- 3. Continuous Carbon Auditing:** Schedule monthly Dark Data Auditor API scans to track data growth and enforce Green Computing compliance.

4. Audited File Inventory Log (Sample Records)

File Identifier	Type	Size	Age	AI Conf.	Automated Action Plan
FILE_0001	.dat	19.14 MB	423d	99.0%	Retain Active Tier
FILE_0002	.zip	116.39 MB	571d	88.9%	Retain Active Tier
FILE_0003	.csv	10.99 MB	216d	90.3%	Migrate to Cold Storage
FILE_0004	.csv	13.54 MB	681d	92.7%	Recommend Deletion
FILE_0005	.dat	1.75 MB	555d	90.4%	Recommend Deletion
FILE_0006	.png	13.55 MB	923d	92.9%	Retain Active Tier
FILE_0007	.log	8.23 MB	762d	96.7%	Recommend Deletion
FILE_0008	.docx	12.41 MB	266d	90.7%	Retain Active Tier
FILE_0009	.mp4	171.19 MB	587d	89.6%	Recommend Deletion
FILE_0010	.xlsx	6.22 MB	903d	91.9%	Retain Active Tier
FILE_0011	.pdf	8.56 MB	197d	90.3%	Retain Active Tier
FILE_0012	.xlsx	3.08 MB	41d	94.1%	Retain Active Tier
FILE_0013	.jpg	7.18 MB	251d	96.1%	Retain Active Tier
FILE_0014	.zip	220.37 MB	798d	93.5%	Retain Active Tier
FILE_0015	.xlsx	7.41 MB	638d	90.1%	Retain Active Tier
FILE_0016	.zip	122.86 MB	842d	96.8%	Recommend Deletion
FILE_0017	.csv	7.63 MB	336d	97.6%	Retain Active Tier
FILE_0018	.dat	5.4 MB	384d	91.7%	Recommend Deletion